

## **Program 10**

Demonstrate minikube dashboard using CLI and Kubernetes Cluster

The program aims to demonstrate how to access and manage a Kubernetes cluster visually using the Minikube Dashboard through the Command Line Interface (CLI).

### **Project Structure:**

```
program_10/  
—>deployment.yaml  
—>services.yaml
```

### **Procedure:**

Create a folder (ex: program\_10)

**Step 1:** Start Minikube and Check the status of Minikube

```
minikube start  
minikube status
```

**Step 2:** Create deployment.yaml file with the following code

```
nano deployment.yaml
```

**Step 3:** Create services.yaml file with the following code

```
nano services.yaml
```

**Step 4:** Apply the files

```
kubectl apply -f deployment.yaml  
kubectl apply -f services.yaml
```

## Step 5: Get pods, deployment, services

kubectl get pods

kubectl get deployment

kubectl get services

```
jeevanc@jeevanc-IdeaPad-Slin-3-15AMN8:~/Devopslab/program10$ kubectl apply -f deployment.yaml
deployment.apps/nginx-deployment created
jeevanc@jeevanc-IdeaPad-Slin-3-15AMN8:~/Devopslab/program10$ kubectl apply -f services.yaml
service/nginx-service created
jeevanc@jeevanc-IdeaPad-Slin-3-15AMN8:~/Devopslab/program10$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
nginx-deployment-6f9664446b-7zwz2   0/1     ContainerCreating   0           17s
nginx-deployment-6f9664446b-hnzz2   0/1     ContainerCreating   0           17s
jeevanc@jeevanc-IdeaPad-Slin-3-15AMN8:~/Devopslab/program10$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
nginx-deployment-6f9664446b-7zwz2   1/1     Running    0           3m52s
nginx-deployment-6f9664446b-hnzz2   1/1     Running    0           3m52s
jeevanc@jeevanc-IdeaPad-Slin-3-15AMN8:~/Devopslab/program10$ kubectl get deployment
NAME          READY   UP-TO-DATE   AVAILABLE   AGE
nginx-deployment  2/2     2             2           4ms
jeevanc@jeevanc-IdeaPad-Slin-3-15AMN8:~/Devopslab/program10$ kubectl get services
NAME          TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)          AGE
kubernetes    ClusterIP   10.96.0.1    <none>         443/TCP           6d22h
nginx-service  NodePort    10.101.108.140 <none>        80:31907/TCP     3m59s
```

## Step 6: Check the minikube services and dashboard

minikube services nginx-service

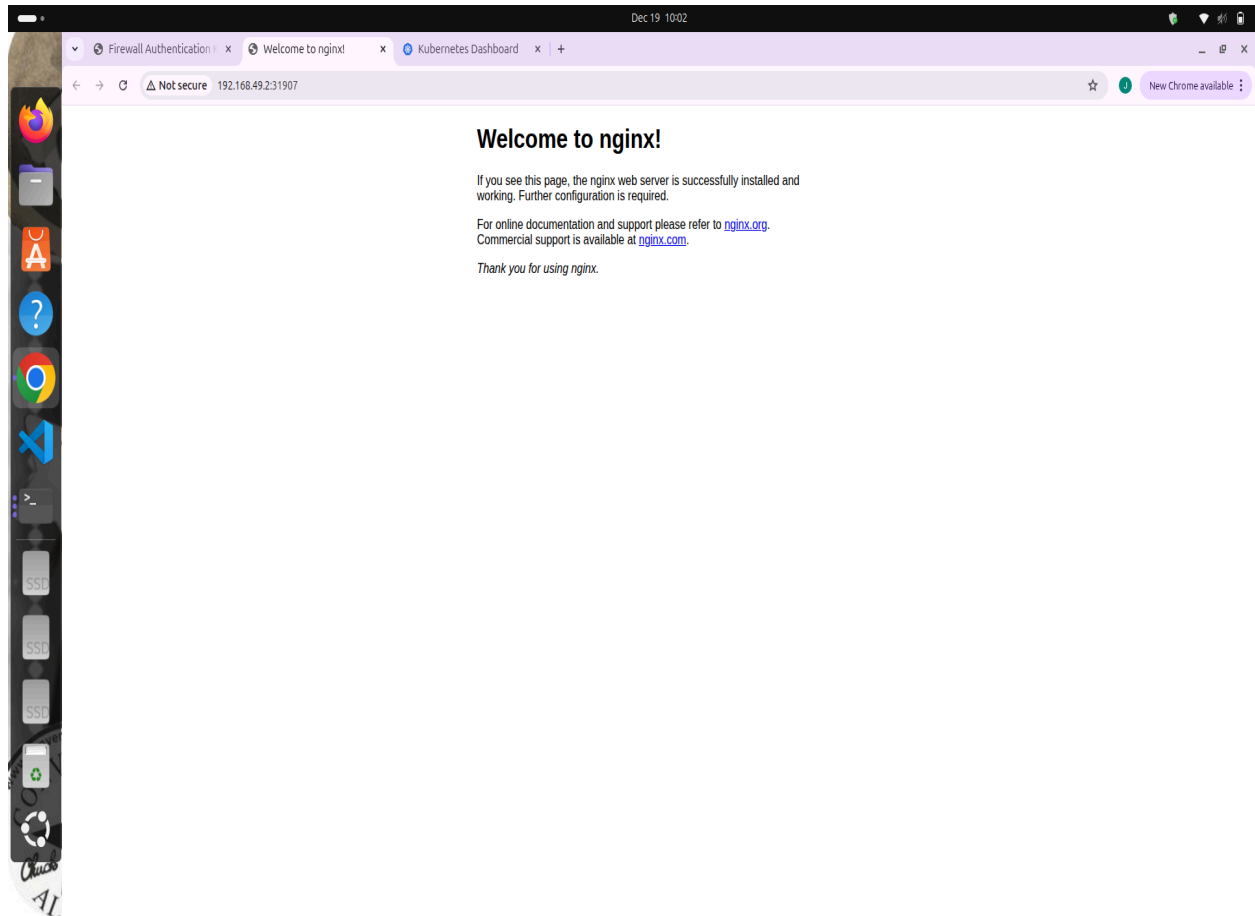
minikube dashboard

```
jeevanc@jeevanc-IdeaPad-Slin-3-15AMN8:~/Devopslab/program10$ minikube service nginx-service
```

| NAMESPACE | NAME          | TARGET PORT | URL                       |
|-----------|---------------|-------------|---------------------------|
| default   | nginx-service | 80          | http://192.168.49.2:31907 |

```
jeevanc@jeevanc-IdeaPad-Slin-3-15AMN8:~/Devopslab/program10$ minikube service nginx-service
Opening service default/nginx-service in default browser...
jeevanc@jeevanc-IdeaPad-Slin-3-15AMN8:~/Devopslab/program10$ minikube dashboard
Opening in existing browser session.
```

## Output:



Dec 19 10:02

Firewall Authentication x Welcome to nginx! x Kubernetes Dashboard x +

127.0.0.1:37021/api/v1/namespaces/kubernetes-dashboard/services/http:kubernetes-dashboard/proxy/#/workloads?namespace=default

new Chrome available

kubernetes

default

Search

+ bell

Workloads

Workloads Cron Jobs Daemon Sets Deployments Jobs Pods Replica Sets Replication Controllers Stateful Sets Service Ingresses Ingress Classes Services Config and Storage Config Maps Persistent Volume Claims Secrets Storage Classes Cluster Cluster Role Bindings Cluster Roles Events Namespaces

Workload Status

Running: 1

Deployments

Running: 2

Pods

Running: 1

Replica Sets

Deployments

| Name             | Images       | Labels     | Pods  | Created        |
|------------------|--------------|------------|-------|----------------|
| nginx-deployment | nginx:latest | app: nginx | 2 / 2 | 21 minutes ago |

Pods

| Name                             | Images       | Labels                                     | Node     | Status  | Restarts | CPU Usage (cores) | Memory Usage (bytes) | Created        |
|----------------------------------|--------------|--------------------------------------------|----------|---------|----------|-------------------|----------------------|----------------|
| nginx-deployment-6f966444b-7zw22 | nginx:latest | app: nginx<br>pod-template-hash: 6f966444b | minikube | Running | 1        | -                 | -                    | 21 minutes ago |
| nginx-deployment-6f966444b-hn2g2 | nginx:latest | app: nginx<br>pod-template-hash: 6f966444b | minikube | Running | 1        | -                 | -                    | 21 minutes ago |